WORKSHEET 8.1

# Average Value of a Function

Unit 8: Applications of Integration

## Key ideas

- The **average value** of $f$ on $[a, b]$ is $\dfrac{1}{b-a}\displaystyle\int_a^b f(x)\,dx$.
- It is the height of the rectangle on $[a, b]$ with the same area as the region under the graph.
- If $f$ is continuous on $[a, b]$, then $f(c)$ equals the average value for at least one $c$ in the interval.
- Do not confuse it with the **average rate of change**, $\dfrac{f(b)-f(a)}{b-a}$, which is the average value of $f'$.

## Worked examples

### Example 1 · A polynomial

Find the average value of $f(x) = 3x^2 - 2x$ on $[1, 4]$.

**Solution.** $\displaystyle\int_1^4 (3x^2 - 2x)\,dx = \Big[x^3 - x^2\Big]_1^4 = 48.$

$\dfrac{48}{3} = 16$**.**

### Example 2 · Where the average is reached

Find the value of $c$ in $[0, 9]$ at which $f(x) = \sqrt{x}$ equals its average value on $[0, 9]$.

**Solution.** $\displaystyle\int_0^9 \sqrt{x}\,dx = 18$, so the average is 2. Solve $\sqrt{c} = 2$.

$c = 4$**.**

### Example 3 · An exponential

Find the average value of $f(x) = e^{2x}$ on $[0, 1]$.

**Solution.** $\dfrac{1}{1}\Big[\tfrac{1}{2}e^{2x}\Big]_0^1$.

$\dfrac{e^2 - 1}{2}$**.**



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## Practice

### CORE

1. Find the average value of $f(x) = 6x$ on $[0, 4]$.

2. Find the average value of $f(x) = x^3$ on $[0, 2]$.

3. Find the average value of $f(x) = \dfrac{1}{x^2}$ on $[1, 4]$.

4. Given $\displaystyle\int_2^7 f(x)\,dx = 30$, find the average value of $f$ on $[2, 7]$.

### EXAM LEVEL

5. Find the average value of $f(x) = x\sqrt{x^2 + 9}$ on $[0, 4]$.

6. The average value of $f$ on $[1, 5]$ is 7. Find $\displaystyle\int_1^5 (f(x) + 2)\,dx$.



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7. Find the value of $c$ in $[0, 6]$ at which $f(x) = x^2 + 2$ equals its average value on $[0, 6]$.

8. The table gives the temperature $H(t)$, in °C, at selected times $t$ hours. Use a trapezoidal sum with three subintervals to estimate the average temperature for $0 \le t \le 10$.

| $t$ | 0 | 2 | 6 | 10 |
|---|---|---|---|---|
| $H(t)$ | 12 | 18 | 20 | 14 |

**Question 9 · Free-response practice**

*Calculator permitted.* The temperature in a greenhouse is $W(t) = 20 + 15\sin\left(\frac{\pi t}{12}\right)$ degrees Celsius, for $0 \le t \le 12$, where $t$ is in hours.

(a) Find the average temperature for $0 \le t \le 12$.

(b) Find the average rate of change of the temperature for $0 \le t \le 6$. Include units.

(c) Find the times at which the temperature equals its average value.



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## Worked solutions

1. $\dfrac{1}{4}(48) = 12.$

2. $\dfrac{1}{2}(4) = 2.$

3. $\dfrac{1}{3}\left(1 - \dfrac{1}{4}\right) = \dfrac{1}{4}.$

4. $\dfrac{30}{5} = 6.$

5. $\dfrac{1}{4}\left[\frac{1}{3}(x^2+9)^{3/2}\right]_0^4 = \dfrac{1}{4} \cdot \dfrac{125-27}{3} = \dfrac{49}{6}.$

6. $\displaystyle\int_1^5 f(x)\,dx = 28$, and $\displaystyle\int_1^5 2\,dx = 8$: total 36.

7. Average $= \dfrac{1}{6}(72+12) = 14$; $c^2 + 2 = 14$ gives $c = 2\sqrt{3}$.

8. $\dfrac{2(12+18)}{2} + \dfrac{4(18+20)}{2} + \dfrac{4(20+14)}{2} = 174$; average $\approx 17.4$ °C.

9. (a) $\dfrac{1}{12}\displaystyle\int_0^{12} W(t)\,dt = 20 + \dfrac{30}{\pi} \approx 29.549$ °C.

   (b) $\dfrac{W(6) - W(0)}{6} = \dfrac{35-20}{6} = 2.5$ °C per hour.

   (c) $\sin\left(\dfrac{\pi t}{12}\right) = \dfrac{2}{\pi}$: $t \approx 2.636$ and $t \approx 9.364$ hours.



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